

1 Task 4 — Heterogeneous Graph Embedding (optional practice)

1.1 Method

The graph. H_{train} is the notebook’s `add_artists(G_train)`, used verbatim: 15,466 nodes (7,624 users + 7,842 artists) and $19,464 + 3,014,361 = 3,033,825$ edges. Every figure in the provided statistics file is reproduced exactly: 7,451 users like at least one artist and 173 like none, while artists per user has median 400 and max 944, matching the table printed in the task description. The artist layer is $155\times$ denser than the social layer, and that single fact drives every result below.

What is reused unchanged. This is what makes the comparison in Table 1 legitimate, so I state it precisely. `train_embedding`, `LinkPredictor` and `CountryClassifier` are *the same Python objects* measured in Task 3: my sampler is a subclass of Task 3’s `WalkEmbedding` that overrides `sample_walks` and nothing else, and Task 3’s module is imported byte-identically (verified by `sha256`). The walk budget (10 walks $\times$ length 40 per node) and every Skip-Gram hyper-parameter ($d=128$, window 5, 5 negatives, 5 epochs, `workers=1` for bit-exact reproducibility) are unchanged. Evaluation is `run_experiment` with H_{train} and $N_{\text{users}} + N_{\text{artists}}$, slicing Z to $[: N_{\text{users}}]$; degree buckets are always taken from the *social* graph G_{train} . **The sampler and the graph are the only things that change.**

Schema convention. A schema is a symmetric type string whose repeating cycle is `schema[:-1]`: U-A-U gives UAUUA..., U-U-A-U gives UUAUUA..., U-U-U-A-U gives UUUUA... At each step the next node is drawn uniformly from the neighbours of the type the schema demands. Since U-A-U-U is a cyclic rotation of U-U-A-U and yields the same multiset of step types, I run one schema per rotation class. The three schemas then form a one-parameter family indexed by the fraction of steps forced to be social — 0, $1/3$, $1/2$, with Task 3’s DeepWalk as the 1 endpoint — which is the axis of Figure 2.

Artist start nodes. `embed_graph` starts walks from *every* node of H , artists included. I rotate the cycle to the start node’s type rather than discarding artist starts, so an artist begins AUUAU... — the same meta-path read from the other end; artists thus get trained vectors that serve as shared context for users, and are sliced off before evaluation. A walk that cannot take its first step (a user with no artist under U-A-U; a socially isolated user under U-U-A-U) is dropped, exactly as Task 3’s sampler returns `[]` for an isolated node, so such users keep an all-zero row in both tasks and the cold-start comparison stays like-for-like. Under U-A-U this gives $(15,466 - 173) \times 10 = 152,930$ walks.

One idea of my own: rare-artist sampling. A uniform $U \rightarrow A$ step lands on a popular artist far more often than a niche one, and a shared taste for a top-100 artist says almost nothing about two users. I therefore add an optional weighting in which a step *into* an artist is drawn with probability $\propto \deg(a)^{-\alpha}$ (steps into a user stay uniform). At $\alpha=1$ this lowers the median popularity of the artist a walk actually visits from 613 to 205 — i.e. to the median popularity of an artist in the dataset, removing the sampling bias towards hubs.

```
class MetaPathWalkEmbedding(WalkEmbedding): # Task 3's class, one method overridden

    def __init__(self, schema="UAU", alpha=None, seed=RANDOM_SEED, **params):
        super().__init__(seed=seed, **params)
        assert schema[0] == schema[-1] # schema must be symmetric
```

```

self.schema, self.cycle, self.alpha = schema, schema[:-1], alpha

def sample_walks(self, G, start):
    tables, ntype = self._typed_tables(G) # node -> {type: (nbrs, cum_weights)}
    t0 = ntype[start]
    if t0 not in self.cycle:
        return []
    k = self.cycle.index(t0)
    rot = self.cycle[k:] + self.cycle[:k] # rotate the cycle to the start's type
    m, L, rng = len(rot), self.walk_length, self._rng

    walks = []
    for _ in range(self.num_walks):
        walk, i = [start], 0
        while len(walk) < L:
            entry = tables[walk[-1]].get(rot[(i + 1) % m])
            if entry is None: # no neighbour of the required type
                break
            nbrs, cum = entry
            walk.append(rng.choice(nbrs) if cum is None # uniform
                        else rng.choices(nbrs, cum_weights=cum, k=1)[0]) # rare-artist
            i += 1
        if len(walk) > 1:
            walks.append(walk)
    return walks

```

Listing 1: The meta-path sampler. Everything else — training, link predictor, country classifier — is unchanged from Task 3.

`_typed_tables` is built once per graph from the notebook’s `typed_adjacency(G)` and cached: for each node it stores the neighbour *list* per type, plus cumulative $\deg(a)^{-\alpha}$ weights when rare-artist sampling is on. Sampling a step is therefore $O(1)$ (or $O(\log d)$ weighted), and the whole $\sim 152,930$ -walk corpus takes 2–8 s against ~ 115 s of Skip-Gram training.

1.2 Protocol, and one control that matters

Every configuration is run at seeds $\{0, 1, 2\}$ and all reported numbers are 3-seed means; the “noise floor” row of Table 1 is the mean within-configuration standard deviation, which is what licenses calling a gap real.

One confound must be removed before any Task 4 number can be compared with a Task 3 number. Task 3 walks from the 6,958 non-isolated *users* (69,580 walks); every Task 4 run walks from all 15,466 nodes of H ($\approx 152,930$ walks), a $2.2\times$ larger Skip-Gram corpus, so a naive reading would credit the artist structure with a gain that is really just more training data. I therefore ran **DeepWalk on the social graph with 22 walks per node** ($6,958 \times 22 = 153,076$ walks, matching the Task 4 corpus to within 0.1%). That is the row Task 4 must actually beat, and it changes the conclusion: the extra budget alone is worth +0.0170 test Hit@1 and +0.0085 test MRR, but nothing on country classification (-0.0004 micro-F1).

1.3 Results

strategy	link prediction				country classification			
	val H@1	val MRR	test H@1	test MRR	val mi-F1	val ma-F1	test mi-F1	test ma-F1
<i>Task 3 — social graph G, 10 walks/node</i>								
DeepWalk (uniform)	0.6040	0.7227	0.6009	0.7207	0.7641	0.6639	0.7629	0.6833
node2vec $p=1, q=0.5$	0.5995	0.7202	0.5959	0.7184	0.7636	0.6621	0.7621	0.6825
node2vec $p=1, q=2$	0.6027	0.7210	0.6011	0.7196	0.7619	0.6467	0.7607	0.6802
node2vec $p=q=0.25$	0.6039	0.7217	0.6061	0.7228	0.7610	0.6581	0.7628	0.6796
RWR $r=0.15$	0.5870	0.7119	0.5889	0.7124	0.7637	0.6521	0.7620	0.6733
deg-corrected $\alpha=1$	0.6003	0.7144	0.5990	0.7121	0.7575	0.6421	0.7607	0.6781
triadic	0.6066	0.7247	0.6087	0.7248	0.7645	0.6833	0.7661	0.6974
<i>Budget-matched control — social graph G, 22 walks/node</i>								
DeepWalk, 22 walks/node	0.6170	0.7298	0.6179	0.7292	0.7623	0.6643	0.7624	0.6852
<i>Task 4 — heterogeneous graph H, 10 walks/node</i>								
type-blind DeepWalk on H	0.4648	0.6306	0.4534	0.6229	0.6898	0.4111	0.6804	0.3794
metapath U-A-U	0.4201	0.5906	0.4133	0.5849	0.6457	0.3630	0.6489	0.3516
metapath U-U-A-U	0.5661	0.6994	0.5525	0.6876	0.7624	0.5738	0.7632	0.5675
metapath U-U-U-A-U	0.6215	0.7428	0.6163	0.7361	0.7956	0.6337	0.7887	0.6353
U-A-U rare-artist $\alpha=1$	0.4270	0.5954	0.4261	0.5942	0.7038	0.4245	0.6857	0.4007
<i>noise floor (mean sd)</i>	0.0038	0.0029	0.0050	0.0033	0.0076	0.0133	0.0038	0.0055

Table 1: The notebook’s `comparison_table()`, Task 3 and Task 4 together. Mean over seeds $\{0, 1, 2\}$; Task 3 rows are the measured Task 3 results, not re-run. The noise floor is the mean within-configuration standard deviation over all 13 rows. Validation columns are the splits on which `fit` selects C and are therefore optimistic; the argument rests on the test columns.

source degree	n	DeepWalk $\times 22$		U-A-U		U-U-A-U		U-U-U-A-U	
		H@1	MRR	H@1	MRR	H@1	MRR	H@1	MRR
0 (cold)	216	0.0494	0.1806	0.3596	0.5320	0.2994	0.4712	0.2824	0.4666
1	266	0.4386	0.5597	0.4098	0.5702	0.4273	0.5924	0.5063	0.6338
2–3	469	0.5736	0.6893	0.4357	0.6026	0.5153	0.6548	0.5465	0.6744
4–7	842	0.6829	0.7818	0.4406	0.6083	0.6065	0.7268	0.6496	0.7606
8–15	939	0.6922	0.7973	0.4441	0.6092	0.6145	0.7371	0.6894	0.7932
≥ 16	1439	0.6644	0.7806	0.3785	0.5601	0.5536	0.6931	0.6423	0.7641

Table 2: Test-split link prediction by the source user’s degree in the social graph G_{train} , 3-seed means. The reference column is the budget-matched control, not the 10-walk DeepWalk. The cold row and every other row point in opposite directions.

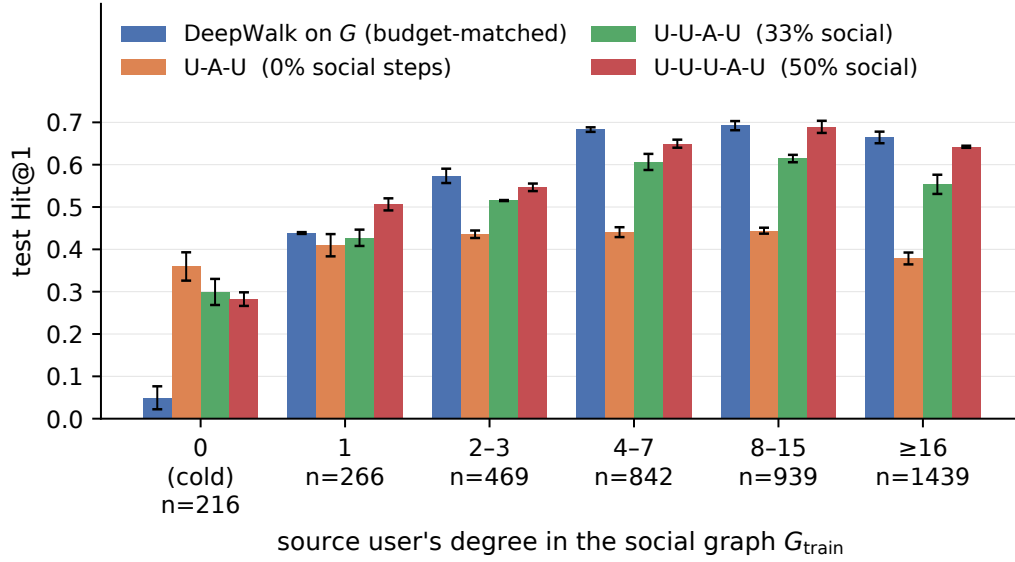


Figure 1: Test Hit@1 by social degree of the source user; error bars are ± 1 sd over three seeds. Adding artists is not a uniform improvement or a uniform regression — it transfers accuracy from the well-connected users to the cold-start ones.

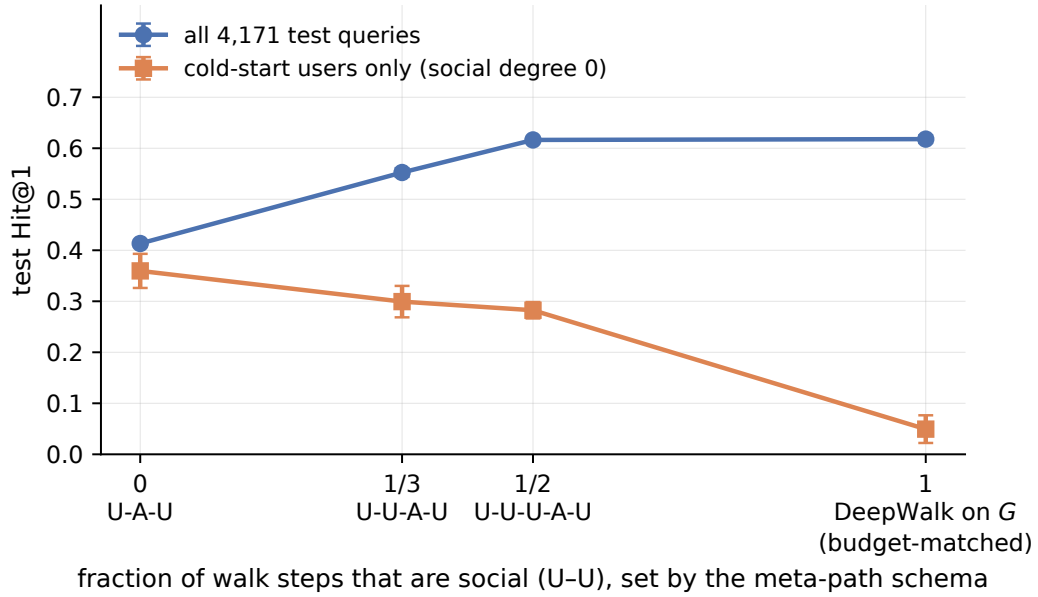


Figure 2: The same three schemas plus the budget-matched DeepWalk, arranged by the fraction of walk steps the schema forces to be social. The two populations move in opposite directions, and the schema is the knob that trades one against the other.

1.4 Analysis

Aggregate link prediction: no, the artists do not help. U-U-U-A-U reaches test Hit@1 0.6163 against the budget-matched control’s 0.6179 — a gap of -0.0016 , or -0.3 noise sd, i.e. nothing. On test MRR it is 0.7361 versus 0.7292, $+0.0070$ or $+2.1$ sd, a marginal edge at best. Compared against the *unmatched* 10-walk DeepWalk the same schema looks like a clean $+0.0154$ Hit@1 win, and that apparent win is almost entirely the larger walk corpus. This is the main reason the control was worth running: without it the honest answer here inverts into a false positive.

Cold-start users: yes, dramatically, and this is not a budget artefact. 666 users have no friend in G_{train} and receive an all-zero row from any purely social walk; 216 test queries have such a source. Their test Hit@1 is 0.0447 for Task 3 DeepWalk and 0.0494 for the budget-matched control — the extra corpus buys them nothing, because no amount of walking reaches a node with no edges. But 647 of those 666 users (97.1%) like at least one artist, with a median of 334 artists each, and 209 of the 216 cold test sources (96.8%) are covered. Through the artist layer they become reachable, and their Hit@1 rises to 0.2824 under U-U-U-A-U and 0.3596 under U-A-U — respectively $5.7\times$ and $7.3\times$ the control, gaps of 8.3 and 11.0 times the 0.0282 mean seed spread of this bucket. This is the one place where the heterogeneous structure does something no amount of social-graph effort can.

Signal or noise? Both, and density decides which. A U-A-U step connects two users who share *any* artist. Sampling 40 users, the median number of others reachable in one such step is 7,307 of 7,623 — **about 96% of the graph** — against a median social 2-hop neighbourhood of 61. A U-A-U edge is therefore almost devoid of information: it is not far from a link to a uniformly random user. That explains the whole shape of Table 2. For a user with ≥ 16 friends the social graph already supplies a sharp, small candidate set, and diluting it with a near-complete graph costs $0.6644 \rightarrow 0.3785$ Hit@1. For a user with 0 friends the alternative is not a sharper signal but an all-zero vector and chance-level ranking, so even a weak ordering — the artist layer is near-complete but not uniform, and how *many* artists two users share still varies — is worth a great deal. Same edges, opposite value, depending entirely on what the user already had.

The meta-path constraint is doing real work. A type-blind DeepWalk on H takes a social step only 0.72% of the time (measured over 31,122 sampled steps), because a user has ~ 395 artist neighbours and ~ 5 social ones; it is *de facto* U-A-U. Yet it beats explicit U-A-U on every column ($+0.0401$ test Hit@1). Those rare social steps are worth a great deal, which is precisely the metapath2vec argument: on a graph this lopsided, leaving the transition rule to the degree distribution throws away the sparse, informative edge type, and the schema is how you take that choice back. The stronger form of the same point is the monotone family in Figure 2: as the forced social fraction rises $0 \rightarrow 1/3 \rightarrow 1/2$, overall test Hit@1 goes $0.4133 \rightarrow 0.5525 \rightarrow 0.6163$ while the cold bucket goes $0.3596 \rightarrow 0.2994 \rightarrow 0.2824$. Notably the cold bucket does *not* collapse to the 0.0494 of a pure social walk even at $1/2$, because although a socially isolated user can never start a U-U-U-A-U walk, the $A \rightarrow U$ step lets other walks reach them.

Country classification: helped on micro-F1, hurt on macro-F1. Against the budget-matched control, U-U-U-A-U gains $+0.0263$ test micro-F1 (0.7887 vs 0.7624, 7.0 sd) and loses -0.0498 test macro-F1 (0.6353 vs 0.6852, 9.1 sd). Both are far outside the noise, and the split between them is interpretable: music taste is regional, so artists carry genuine country signal and lift the large classes that micro-F1 measures — but that signal is coarse, regional rather than

national, while macro-F1 is dominated by the small countries, which in the social graph are tight, well-separated communities that the near-complete artist layer blurs together. Task 3’s triadic walk remains the best model in the study for macro-F1 (0.6974) and no heterogeneous variant comes close.

Rare-artist sampling works, and confirms the diagnosis. Weighting $U \rightarrow A$ by $\deg(a)^{-1}$ improves plain U-A-U on all eight columns: +0.0129 test Hit@1 (2.6 sd), +0.0094 test MRR (2.9 sd), +0.0368 test micro-F1 (9.7 sd) and +0.0491 test macro-F1 (9.0 sd). Since the only change is which artist a walk passes through, this is direct evidence that it is specifically the *popular* artists that inject the noise, and that the density problem is partly fixable inside the sampler. It does not rescue U-A-U to competitiveness — the schema is still missing the social edges that matter most — but it is the largest single improvement I obtained over a baseline schema.

1.5 Conclusion

Heterogeneous structure helped *one* of the two downstream tasks and one sub-population, and hurt elsewhere. It did essentially nothing for aggregate friend recommendation once the walk budget was matched; it improved country micro-F1 by a wide margin and degraded macro-F1 by a wider one; and it improved cold-start friend recommendation by roughly $7\times$, which is the result worth keeping. The practical recommendation that follows is not a single best schema but a split one: use a social-graph walk (Task 3’s triadic) for users who have friends, and a U-A-U walk — ideally with rare-artist weighting — for the 9% of users who do not. Table 2 says the crossover sits between degree 0 and degree 1. The broader lesson is that on a graph where one edge type is $155\times$ denser than the other, the meta-path is not a modelling flourish but the only thing standing between the walk and a near-complete graph.